A Limited Multiarticle Example of AMLD Settlement

Trading to Modus Ponens, Quotient Hunting, and Informal Basin Escape
A chess-style "White Wins" interpretation with proof-carrying return

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Research program and original conceptual framework
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See the AI disclaimer that follows this essay.

Status. A *restricted synthetic example* with paper-level proofs and locally executed finite checks. It does *not* certify the general Multiarticle speedup conjecture, the original research papers, a running AMLD federation, a solution to J4, or any chess position.

Abstract

A useful way to study automated metalogical settlement is to make a proof-search system change its representation while preserving the accepted proof. This article builds an explicit infinite family illustrating that idea. An original calculus with modus ponens and three report-extension rules is traded to a consequence-equivalent MP-only calculus; a small proof-bearing two-island interface gives report-parity invariants; a Quotient Hunter selects a labelled, settlement-preserving quotient; and an accepted abstract route is lifted into an original-calculus certificate. For words of length at most n over three symbols the history graph has $(3^{n+1}-1)/2$ states, but the reachable quotient has $4n$ states. A local step-by-step rule checker, distinct from the proof-generation routine, accepts the displayed MP-only and returned proofs for seven values of n; the infinite-family claims receive ordinary mathematical proofs, not formal-assistant receipts. An auxiliary alternating reachability game clarifies the chess analogy: "White wins" means a strategy that forces a certified accepting state against all legal Black choices, not the existence of one good path. We relate the construction, with explicit provenance limits, to AMLD's six settlement outcomes, Multiarticle proof hypergraphs, GBG, MUSIC, SYMPHONY, TISO, theoremgraph reversal, and informal basin escape. Polynomial fully charged construction is established within a declared symbolic model; an intrinsic speedup against an equal-resource optimal comparator is *not* claimed.

Keywords. Automated metalogical deciders; AMLD+0; proof certification; trading theorem; modus ponens; Multiarticle Theory; Quotient Hunter; generalized board games; dreaming; symphonies; basin escape; reachability games.

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1 What the example can, and cannot, say

The underlying research program asks whether cooperating or single automated provers can find a better representation of the same settlement problem. A representation may change proof length, search branching, or the cost of producing a certificate without changing the underlying consequence relation. This is the distinction at the heart of the Trading Theorem and the Quotient Hunter [2]. The MP-only return paper provides a richer first-order version of the trading step under explicit quantifier-scope assumptions [3]. The proof-bearing Multiarticle program insists that an article union retain proof edges, typing, hypothesis scopes, and provenance rather than merely concatenate conclusions [4, 5].

This article deliberately supplies a *limited worked model*. Its two theorem islands are newly defined parity contracts, not extracted and independently validated theorems from the five cited research manuscripts. It illustrates a plausible interface to their theories; it does not establish that those theories are computationally indispensable to the toy result. The synthetic character of the example is crucial to a defensible originality and certificate claim.

Two levels of "certified" must also be distinguished. In a *formal model*, a proof object is certified when it passes the named finite checker and satisfies the model's stated trust contract. In a *research-publication claim*, independent machine certification of an infinite-family theorem would require a separately trusted implementation and accepted parametric proof artifact. We have the first, locally and for finite instances, but not the second [1, 9].

2 A map of the participating theories

Theory or mechanism	Role in the present exposition	Epistemic status here
AMLD+0/AMLD+000	Player Zero coordinates proof, refutation, dependence, independence, and consistency checks; the original verifier controls acceptance.	Protocol-level, offline emulated
Trading Theorem; MP-only return	Replace report rules by closed implication resources; recompile accepted proofs into the frozen original calculus.	Proved in a scoped toy calculus
Multiarticle	Join two typed source contracts with an explicitly checked acceptance bridge and distinct source labels.	Synthetic, proof-bearing instance
GBG and theoremgraph	Search is a labelled legal-move game; reversed MP obligations are planning objects until discharged.	Formal example/interpretation
MUSIC	Individual report histories generate possible dreams; forgetting the precise history can reveal a useful continuation quotient.	Motivational analogy
SYMPHONY	Coordinated legal moves require provenance and causal order, unlike an arbitrary set of possible dreams.	Motivational analogy
Quotient Hunter	Tests candidate state summaries and retains the first accepting, transition-respecting key.	Executed finite model
TISO and resource accounting	Exposes why allocation, communication, expansion, and checking must be charged before claiming a speedup.	Accounting requirement
J4, DNI, attained labels	Motivating distant applications and cautions about undecidability and physical claims.	No result claimed

Table 1: Theories invoked, and the difference between a proved toy instantiation and a conceptual connection.

2.1 AMLD is a settler, not merely an ATP

A singleton proof searcher asks whether it can construct a proof. AMLD-class settlement instead distinguishes six possible *reports*: P (proved), R (refuted), D (dependence), I (independence), C (contradiction), and U (unknown). The board functions are Player Zero, P, R, D, I, and C; U is the residual outcome, not a seventh board worker. Correctness requires separate proof standards for each outcome. In particular, failing to find a proof does not establish R or I. The semantic-bus bootstrap forbids promoting unverified hints to the certified BANK [1]. This article's finite checker is a local research artifact, not evidence of six actually executing agents.

2.2 What is a dream, a symphony, and a zip?

In the MUSIC framework, a dream is a union of location reports, D_c is the set of possible dreams from condition c, and $\mathrm{MUSIC}(c) = \mathcal{P}(D_c)$ is the family of possible dream collections [7, 6]. A SYMPHONY additionally records legally executable cooperative histories, transfer authorization, causality, and verification. Equal MUSIC possibility sets need not imply equal reachable certified symphonies [8, 6, 4].

A Multiarticle zip in the source papers is a typed proof-hypergraph operation, not merely a union of words: vertices carry claims and contexts; edges retain premises, checking contracts, proof artifacts, costs, and provenance. The six low-fan-in shells in one paper count certified bridge applications, while the +1 to +6 entries in another paper track premise-support size; these two interpretations must not be silently identified [4, 5].

3 The frozen symbolic family

Definition 3.1 (Report-history task)**.** Fix $n \geq 2$, written in unary. Let $\Sigma = \{0, 1, 2\}$ and

$$W_n = \{w \in \Sigma^* : |w| \leq n\}.$$

The initial state is the empty history ε. A move labelled $c \in \Sigma$ appends c whenever $|w| < n$. Let

$$z(w) = \#_0(w) \bmod 2, \qquad o(w) = \#_1(w) \bmod 2.$$

A history accepts if $\mathrm{Acc}_n(w)$ holds, where

$$\mathrm{Acc}_n(w) \iff |w| = n \ \wedge \ z(w) = 0 \ \wedge \ o(w) = 0.$$

The original checker accepts the associated formal proof of the goal atom G_n only from authorized start, move, and terminal-acceptance resources, specified in Section 6.

A report value 2 is neutral for both parity conditions. Thus 2^n is a concrete accepting history for every n. This fact is useful for a short constructive witness, but also limits performance claims: a strong equal-information comparator can see the same simple witness.

Lemma 3.2 (History count)**.** *The explicit history tree contains*

$$|W_n| = \sum_{j=0}^{n} 3^j = \frac{3^{n+1} - 1}{2}$$

states. No assumption that all must be enumerated in order to settle G_n is warranted.

Proof. There are 3^j words of length j. Summing the finite geometric series gives the formula. □

4 A two-island Multiarticle and its proof-bearing seam

Call the first synthetic island $\mathcal{A}$ and the second $\mathcal{B}$. The terminology "island" is structural: it does not claim either is an authenticated export of a named published source.

$$\mathcal{A}: \quad z(\varepsilon) = 0, \quad z(w0) = z(w) \oplus 1, \quad z(w1) = z(w2) = z(w), \tag{1}$$

$$\mathcal{B}: \quad o(\varepsilon) = 0, \quad o(w1) = o(w) \oplus 1, \quad o(w0) = o(w2) = o(w). \tag{2}$$

Each recurrence follows by structural induction from its corresponding count. A small, typed seam pairs the updates for the same word and report label. A cross-island conclusion then states that acceptance at the terminal depth requires *both* zero parities.

Definition 4.1 (Proof-bearing toy zip)**.** Let a source record be a tuple $(\phi, \Gamma, L, V, s, \pi, \omega)$ recording a formula, permitted assumptions, typing context, verifier identity, check status, derivation, and originating island. The toy zip is the tagged proof union of $\mathcal{A}$ and $\mathcal{B}$ plus the common report-word interface and the binary-fan-in bridge

$$\big[|w| = n \wedge z(w) = 0\big], \quad [o(w) = 0] \quad \Longrightarrow \quad \mathrm{Acc}_n(w).$$

The bridge conclusion is checked against the frozen acceptance definition; no new unproved object-level axiom is added to the original calculus.

The bridge is inexpensive and genuinely uses two separately labelled facts *within this toy source specification.* But if a system is given the original word, it can compute both counts itself. Thus this is a source-interface illustration, not an absolute theorem that either publication is indispensable.

Proposition 4.2 (Restricted source-coordinate necessity)**.** *Among projections of the form* $(|w|)$*,* $(|w|, z(w))$*,* $(|w|, o(w))$*, and* $(|w|, z(w), o(w))$*, none of the first three preserves acceptance for all terminal words, while the last does.*

Proof. At depth n, the word 2^n accepts. The word $0\,2^{n-1}$ has the same depth and o-parity but fails the z-condition. The word $1\,2^{n-1}$ has the same depth and z-parity but fails the o-condition. Hence every projection omitting either parity merges an accepting history with a rejecting one. The full triple determines the acceptance predicate by definition. □

This is a useful source-removal *test* in the declared candidate grammar. It does not show that a clever different representation, already equipped with the entire report history, could not reconstruct the omitted parity.

5 The Quotient Hunter and exact settlement congruence

A symbolic Quotient Hunter tests the four candidate signatures from the preceding proposition, starting with the smallest. Each proposed signature is checked for acceptance uniformity and labelled successor compatibility. The first three fail by explicit terminal witnesses; the fourth succeeds. The search takes constant-many schematic trials and does not enumerate all 3^n histories.

Theorem 5.1 (Exact finite-history quotient)**.** *For every* $n \geq 2$*,*

$$K_n(w) = \big(|w|, z(w), o(w)\big)$$

defines a labelled settlement congruence on W_n*. Every abstract labelled transition lifts from every concrete representative. The number of reachable quotient states is exactly* $4n$*.*

Proof. If $K_n(u) = K_n(v)$, then u, v have equal depth and parity. If this depth is n, both have identical acceptance status and no further moves. Otherwise each report c is enabled from both and updates the key by

$$(t, a, b) \xrightarrow{c} (t + 1, a \oplus [c = 0], b \oplus [c = 1]).$$

Therefore corresponding labelled successors have identical keys. Conversely, any such transition lifts from any representative u by appending the same c; terminal acceptance is uniform on each key class. This proves a strong labelled congruence and constructive lifting.

At depth zero only $(0,0)$ is reachable; at depth one the three parity pairs $(1,0),(0,1),(0,0)$ are reachable; at each depth $t \geq 2$ all four pairs are reachable (choose up to two nonneutral symbols and pad with 2). Thus $1 + 3 + 4(n-1) = 4n$. □

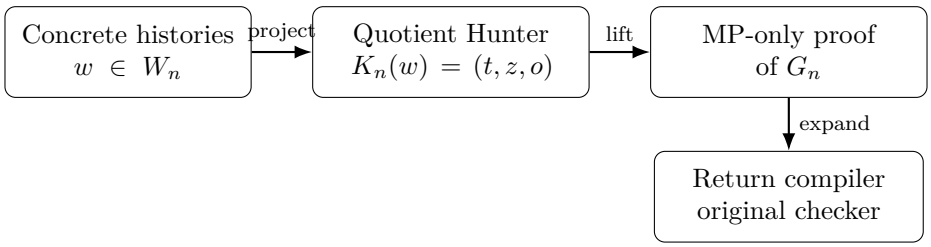


Figure 1: The finite illustrative chain. The sources $\mathcal{A}$ and $\mathcal{B}$ justify the quotient; only a checked original proof may be admitted.

The power of this quotient is a representation result: a history tree with exponentially many nodes is reduced to $4n$ reachable classes. It does not imply that every proof search using the tree takes exponential time, or that the Quotient Hunter beats an optimal comparator capable of using the same key [2, 6].

6 Trading to modus ponens and returning to the original verifier

To avoid hiding the first-order generalization side conditions, the worked certificate uses a simpler, explicit ground propositional language. The separate first-order MP-normalization theorem in [3] is a source of method, not an unverified step inside this example.

Definition 6.1 (Original and traded presentations)**.** Let the language contain atoms $H(w)$ for $w \in W_n$ and one goal atom G_n, with implication $\to$. Both presentations use a standard sound implicational Hilbert base containing the instance $\beta \to (\alpha \to \beta)$, the start axiom $H(\varepsilon)$, and terminal axioms

$$H(w) \to G_n \quad \text{for } |w| = n,\ z(w) = o(w) = 0.$$

The original presentation F_n has MP plus the three proper report rules

$$r_c : \quad \frac{H(w)}{H(wc)} \qquad (|w| < n,\ c \in \Sigma).$$

The traded presentation $F_n^\star$ has MP as its *sole* proper inference rule and adds the instance schema

$$E_{w,c} :\ H(w) \to H(wc) \qquad (|w| < n,\ c \in \Sigma).$$

The underlying language, formula meanings, start assumptions, terminal condition, and frozen checker V_n for the original presentation do not change during a search or return translation.

Theorem 6.2 (Exact trade for every hypothesis bank)**.** *For every additional set of formulas* Γ *and every formula* θ *of the shared language,*

$$\Gamma \vdash_{F_n} \theta \quad \Longleftrightarrow \quad \Gamma \vdash_{F_n^\star} \theta.$$

Both directions have finite proof-producing translations preserving Γ.

Proof. For the forward translation, replace every use of r_c by the instance $E_{w,c}$ followed by one MP step. Other proof occurrences are unchanged, so no additional open assumption is introduced.

For the reverse translation it suffices to show that every $E_{w,c}$ is a theorem of F_n, under *empty* open-assumption support. Starting from the axiom $H(\varepsilon)$, apply the primitive report rules along the letters of wc to obtain $H(wc)$ in a finite closed proof. The Hilbert instance $H(wc) \to (H(w) \to H(wc))$ then yields $E_{w,c}$ by MP. Replace each traded axiom occurrence by that closed original proof and retain all existing MP steps. Since every substituted proof is closed, it is admissible inside any original Γ-context. This is the contextual mutual-reproducibility mechanism of [2] in a particularly explicit form. □

Remark 6.3. This trade is deliberately tailored: every $H(w)$ is already a closed theorem of F_n. The proof does *not* say arbitrary contextual rules can be replaced by implication axioms, nor that trading improves proof complexity. The first-order study [3] carefully distinguishes a scoped quantifier rule from the invalid unrestricted implication $P(x) \to \forall x P(x)$.

Proposition 6.4 (Finite proof and return size)**.** *For the witness* $w = 2^n$*, the MP-only calculus has a proof of* G_n *with* $2n + 3$ *lines. Expanding its report-edge axiom occurrences with the explicit closed proofs in the preceding theorem gives an original-calculus proof with* $3 + n(n + 1)/2 + 4n$ *lines under the declared tree-expansion serialization convention.*

Proof. The traded proof begins with $H(\varepsilon)$, then for each of the n reports uses one $E_{2^j,2}$ axiom and one MP step, and finishes with one accepting terminal implication and one MP step. Hence $1 + 2n + 2 = 2n + 3$ lines. In the return trace, the jth report edge is replaced by a closed derivation of the corresponding $H(2^{j+1})$, which contains one start line, $j + 1$ report-rule lines, one Hilbert K-instance, and one MP step. Summing these expansions, plus the unchanged lines and terminal check, yields the displayed count, as in the deterministic audit code [9]. □

7 Theoremgraphs, proof obligations, and the BANK gate

A reversed MP dependency illustrates the difference between *discovery* and *certification*. To seek G_n backwards, the algorithm may conjecture a predecessor $H(w)$ and the implication $H(w) \to G_n$. The reversed AND-node records two obligations; it does not make either obligation true. Only after each premise is proved under authorized scope, and the whole MP DAG is replayed, may it be translated back to F_n and submitted to the original checker [3].

AMLD's Player Zero can schedule competing candidate theoremgraphs, Quotient Hunter experiments, and source-seam audits. P proposes checked proof objects; R tests candidate refutations or bad lifts; D tracks conditional bridges and imports; I distinguishes true independence from failed search; C tests consistency and scope; no component silently promotes an unverified hint to BANK [1]. This is a method of organising mathematical work, not evidence that the named modules are separately executing programs in the present article.

8 Chess-style "White wins" and informal basin escape

The phrase "White wins" has a precise possible interpretation in an appropriate *two-player reachability game*. It is not, by itself, a proof that a particular chess position is winning, nor is a single successful path sufficient for an adversarial winning claim.

Definition 8.1 (A derived challenge–response GBG)**.** For $m \geq 1$, use a report-history task of maximum length $2m$. In round i, Black first chooses a challenge $b \in \{0, 1\}$ and appends b. White observes b, then chooses any reply $c \in \Sigma$ and appends c. After m rounds, White wins exactly when the recorded report history has both parities even and a valid checked proof of G_{2m}. The goal is reached against a Black strategy τ if the White strategy σ forces the terminal condition. Formally,

$$\mathrm{Win}(s_0) \quad \Longleftrightarrow \quad \exists\sigma\ \forall\tau:\ \mathrm{Run}(s_0, \sigma, \tau) \text{ reaches a verified White-win terminal.}$$

Theorem 8.2 (A compact certified winning-strategy argument)**.** *In this derived game, White has a positional response strategy $\sigma(b) = b$ that reaches a legal accepting terminal against every legal Black sequence. The associated concrete terminal histories have expandable MP-only proofs and accepted original-calculus return proofs.*

Proof. At the start of each round the two parity counters are $(0, 0)$. If Black chooses b, one parity toggles; White appends the same b, so that parity toggles back. The other parity is unchanged. By induction on the number of rounds, after m rounds both parities are zero and the history has length $2m$. All report steps are legal, and the terminal implication for that particular history is an allowed axiom instance. The finite MP-only path proof and the closed-expansion return translation from Section 6 produce an original-calculus proof of G_{2m}. The inductive strategy argument is uniform in Black's choice and need not enumerate Black's 2^m possible challenge sequences. □

Remark 8.3 (Why this is not a solution of chess). The construction is an artificial generalized board game inspired by the *logical form* of a winning-strategy assertion. Real chess includes different legal actions, captures, checks, repetition and terminal conventions. The claim $\exists\sigma\forall\tau$ must be interpreted with the actual game's rules before it becomes a statement about chess. Here "White wins" means a verified objective inside this one defined GBG [10, 5].

8.1 The informal basin metaphor

In heuristic-search language, a *basin* can describe a region where a search policy repeatedly expends work on histories that differ syntactically but have the same relevant future. This usage is *informal*: our model does not define or prove a dynamical-systems basin, an attractor, or an escape-time theorem.

The history family provides an illuminating counting fact. The quotient has $4n$ reachable classes but the concrete family has $(3^{n+1} - 1)/2$ words. Hence some quotient fiber contains at least

$$\frac{3^{n+1} - 1}{8n}$$

concrete histories. A history-oriented explorer may spend effort revisiting that fiber. A Quotient Hunter can replace such indistinguishable histories by their shared abstract state, changing the navigation geometry without changing acceptance. *This* is the limited sense in which the example depicts "escaping a basin." It does not prove that a strong comparator must first enter such a basin or that every AMLD search benefits.

The chess-style game offers a second perspective: searching for one happy path ($\exists$ a line) is weaker than certifying a strategy ($\exists\sigma\forall\tau$). The proven mirroring strategy collapses an exponentially branching response tree into a small invariant. In AMLD terminology, a proof of the relevant winning condition must still pass an explicit verifier and may not be inferred from an intuitive board visualization.

9 Fully charged costs and honest comparisons

A concrete claim of computational advantage must charge at least

$$\begin{aligned} C_{\text{end}} = {} & C_{\text{trade}} + C_{\text{source}} + C_{\text{zip}} + C_{\text{hunt}} + C_{\text{construct}} \\ & + C_{\text{search}} + C_{\text{lift}} + C_{\text{return}} + C_{\text{verify}} + C_{\text{communication}}. \end{aligned} \tag{3}$$

This bookkeeping follows the separation of compressed/search cost from expanded verifier cost emphasized in the Trading studies, the symphonic-compression limitations, and the TISO-oriented Multiarticle discussions [2, 3, 6, 5].

Proposition 9.1 (Polynomial end-to-end synthetic construction)**.** *In the uniform symbolic implementation described here, given unary n, the construction of the two parity contracts, four candidate quotient checks, explicit reachable quotient table, one accepting MP-only proof, its closed return translation, and a direct local proof check can be completed within $O(n^3)$ elementary symbol/bit operations. This upper bound includes proof reconstruction and checker scanning, and excludes no necessary exhaustive search over W_n.*

Proof. The two recurrences and the candidate grammar have constant-size symbolic descriptions. The transition/acceptance checks operate on four parity pairs and three labels, with unary depth parameter, giving at most polynomial work. The reachable quotient has $4n$ states, each representable with $O(\log n)$ index bits. The MP-only certificate has $2n + 3$ lines; the returned certificate has $O(n^2)$ lines, each displaying at most $O(n)$ report symbols, so its naïve serialization and line-by-line verification cost $O(n^3)$ bit operations. Formula-schema membership, edge legality, and assumption-scope checks are explicit and polynomial on the displayed symbols. Additional small orchestration and communication records can likewise be charged within the same polynomial bound for this single-process synthetic instance. □

Remark 9.2. The bound is a sufficient construction bound, not a claim of optimality. It does not establish a speedup over a serious equal-information comparator: such a comparator can emit the visibly accepting word 2^n and check its direct proof in polynomial time without hunting any quotient. An optimal simulator that can run the quotient algorithm at equal cost cannot be strictly outperformed merely because its target is called "centralized." This is the central limit identified in [6, 9].

9.1 A portfolio guarantee that survives a matched comparator

A modest positive operational guarantee remains. If a baseline algorithm and the quotient method each take a known-to-the-analysis finite number C_B, C_Q of *charged* steps to output a fully verified certificate, a single processor may alternate one step of each. With bounded switch overhead s, a completed certificate appears within

$$C_{\text{portfolio}} \;\le\; 2(1+s)\min\{C_B, C_Q\} + O(1),$$

assuming both are resumable and their initialization and final verification are included in their own step counts. The comparator may itself use quotient hunting. This is a portfolio bound, not strict superiority over that comparator [2, 9].

10 What the deterministic finite checker establishes

The companion offline script `AMLD000_Relaxed_Trading_QH_Verifier.py` accepts candidate MP-only steps, expands every report-edge axiom into an original-rule proof, and checks the returned trace. It also checks the twelve local parity-update cases and tests candidate projections. It rejected three intentionally malformed certificates: forged MP, an invalid terminal axiom, and an illegal report edge. No paid APIs or external model agents are involved [9].

A separate companion strategy test enumerated every Black challenge sequence for $m = 1, 2, \ldots, 7$ rounds (254 sequences altogether), applied White's mirroring response, generated the corresponding MP-only proof, and had the original-rule checker accept every returned proof. The associated code and JSON results are included in the reproducibility bundle. This is finite corroboration of the inductive game theorem in Section 8, not a substitute for its uniform proof.

n	$\lvert W_n\rvert$	Reachable $\lvert Q_n\rvert$	Returned proof lines
2	13	8	14
4	121	16	29
8	9,841	32	71
16	64,570,081	64	203
32	2,779,530,283,277,761	128	659

Table 2: Exact formulas evaluated at selected parameter values. The local checker was executed at $n = 2, 3, 4, 6, 10, 16, 32$; $n = 8$ is included here as a formula value, not as an executed checker case.

These are finite tests plus an ordinary induction proof. The local Python implementation has not been independently audited as a foundational proof kernel. It does not prove the actual imported article seams, MP-normalization for arbitrary first-order theories, the existence of a physical J4 basin escape, or an intrinsic advantage over equally powerful algorithms.

11 AMLD+000 three-pass interpretation and settlement ledger

The preceding mathematical development can be organized in three AMLD-style passes without claiming that distinct named agents ran autonomously.

Pass 1. Construction (P and Player Zero). Freeze F_n, G_n, V_n; build the two source contracts; form the typed bridge; find the full parity quotient; construct an MP-only proof and return it to V_n. The proved restricted properties are consequence equivalence, exact congruence, and polynomial end-to-end construction.

Pass 2. Adversarial audit (R, D and C). Reject one-parity projections with terminal counterexamples, distinguish sound proof lifting from mere Boolean reachability, detect fake checker receipts, and charge decoded proof length. A matched optimal comparator may copy the method, defeating any claim of inherent competitive superiority.

Pass 3. Verification and status (P, I, C, Player Zero). Finite proof objects pass a local executable checker; the uniform generalization is supported by a paper-level inductive argument. No independence proof, formally certified all-source zipper, or externally accepted general-purpose theorem certificate has been produced.

Label	Appropriate conclusion
P	Scoped paper-level proof of the synthetic quotient, MP trade, return translation, and small "White wins" strategy.
R	Conditional impossibility of a strict advantage over an optimal, exact cost-preserving simulator.
D	Stronger claims depend on authenticated source imports, formal checker contracts, and the comparator model.
I	No independence certificate; absence of proof is not independence.
C	No inconsistency of the frozen synthetic premises found; this is not a global consistency proof.
U	The original strong Multiarticle performance conjecture remains unresolved.

Table 3: Outcome labels and what the available evidence licenses.

12 What remains open: the actual Multiarticle basin-escape theorem

The present construction is not an empirical victory for AMLD over a strong ATP. It is an *existence illustration* for representation-safe settlement. To turn it into the larger research theorem, at least the following obligations must be met:

(i) Import actual typed statements and proof artifacts from the named GBG, MUSIC, SYMPHONY, TISO, Trading, and MP-only manuscripts, with scope-correct interface contracts. A semantic association is not a machine-checked zipper.

(ii) Produce a discovered invariant not already cheaply obtainable by a matched quotient-capable comparator. Show source-removal necessity in an explicit competitor model, not merely the need to retain two parity coordinates.

(iii) Supply an independently trusted parametric checker, including back-translation to the original verifier and all communication and proof-expansion charges.

(iv) Prove a nontrivial lower bound for a *predeclared competitive algorithm* with the same information and allowed operations, or identify the exact architectural limitation responsible for the advantage. Do not require a strict advantage over an optimal class containing the proposed algorithm.

(v) In proposed physics applications, retain the actual gravitational and quantum equations and their source constraints. A toy GBG strategy or scheduler improvement does not settle J4 or bridge general relativity and quantum mechanics [5].

It is useful to name the aspiration: *certified Multiarticle basin escape* would mean that a proof-bearing cross-island invariant helps a settler leave a demonstrably expensive search region, reach a verified settlement, and return a proof to the frozen checker, all with audited costs. Our model makes this description exact in a small setting, while leaving the scientifically consequential speedup open.

13 Conclusion

A history tree can look enormous while a small invariant controls its legal continuation and settlement status. Here two typed parity contracts, a symbolic Quotient Hunter, a consequence-preserving MP-only trade, and an explicit return compiler yield an infinite family with an exponential-to-linear state compression and polynomial certificate reconstruction. The companion two-player GBG turns the informal slogan "White wins" into an actual $\exists\sigma\forall\tau$ strategy assertion with a short inductive proof.

For AMLD, this limited example illustrates the difference between *finding a promising escape*, *proving that the escape preserves settlement*, and *winning a fully charged competition*. The first two receive scoped positive mathematical arguments here. The third, for the actual cross-article research program, remains an open theorem.

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